Diagnosing Prototype Estimation in Few-Shot Class-Incremental Learning: Training-Free Geometry Selection and Unlabeled-Query Prototype Rectification

Md Tanvir Hossain Saikat* Md Akib Haider* Md Hasanul Kabir
Islamic University of Technology
{tanvirhossain3, akibhaider, hasanul}@iut-dhaka.edu

Abstract

Frozen visual encoders make prototype heads competitive in few-shot class-incremental learning (FSCIL), but it remains unclear how much of the residual head gap is sensitive to noisy novel-class means rather than to the subsequent changes introduced by classifier refitting. On identical features, we construct a retrospective matched counterfactual: first replace each five-shot novel mean with its complete-data counterpart under fixed cosine scoring, then fit a linear probe. Across sixteen strong encoder–dataset pairs, mean replacement accounts for the majority of this ordered gap. Shot and classwise diagnostics support a finite-sample mechanism, whereas a weaker base-only encoder does not satisfy the same criterion. This diagnosis motivates ShrinkSel, a support-only selector over prototype radius and covariance shrinkage whose family contains shared LDA as an exact fallback. Across sixteen evaluated pairs, it improves LDA on fifteen by 0.04–1.52 *points and selects the exact LDA endpoint on Aircraft. We further introduce Query Prototype Rectification (*QPR*), a batch-transductive extension that uses gated unlabeled queries to update novel directions while preserving the selected geometry. On CLIP ViT-B/16, QPR adds* 1.21–2.03 *points; exact query-wise self-exclusion retains the benefit, showing that the gains do not require self-inclusion. The update repairs scarce novel prototypes at a measurable base-class cost and weakens with plentiful supports or very small cohorts. Thus the evidence available for prototype estimation is a central constraint in frozen-feature FSCIL.*

1. Introduction

Few-shot class-incremental learning (FSCIL) must add new classes from a handful of labels without losing the base classes [27, 39]. Most methods address representation drift or catastrophic forgetting [26, 37, 38, 40]. Freezing a strong encoder prevents representation drift and parameter overwriting, yet all-seen performance can still decline through class competition. The resulting head is asymmetric: base classes have reliable statistics, while each novel class is represented by a noisy few-shot mean. We ask how much of the tested head gap is sensitive to this estimation error before a more flexible classifier is fitted.

The answer depends on the information budget. As Fig. 1 shows, complete novel-class splits are used only for retrospective analysis. A *support-only* predictor may use frozen image features, base statistics, and labeled supports; a *query-assisted* predictor may additionally aggregate an unlabeled test cohort. Neither deployable path uses complete novel labels, text, or a known query-class marginal.

We first construct an ordered matched counterfactual path: replace only the five-shot novel means under cosine scoring, then fit a linear probe to the same complete training features. Across sixteen strong CLIP, MAE-ft, and supervised ResNet settings, the first replacement accounts for most of this ordered gap. Increasing the shot count, measuring decision repairs, and relating prototype direction error to classwise loss support the finite-sample interpretation. Because a weaker base-only ResNet control does not show the same majority pattern, this is a bounded result for the tested representations and accounting order, not a universal law of FSCIL.

The diagnosis suggests correcting geometry without fitting a fragile novel-class head. ShrinkSel selects among unit, raw, and bias-corrected prototype radii jointly with a pooled-covariance shrinkage path. Selection and a safety check use class-disjoint base episodes; cosine and shared-covariance LDA are endpoints, so the method can fall back exactly to LDA. When an unlabeled cohort is permitted, our new QPR operator admits confident novel-like queries and updates only novel directions, retaining the support-selected radius and metric. Query-wise self-exclusion tests whether the evaluated query itself is responsible for the update.

Our contributions are:

- a matched mean-versus-rule counterfactual path, with crossed uncertainty and empirical checks of the finite-shot mechanism;
- ShrinkSel, a base-selected support-only geometry family

*Equal Contribution.

with a conservative LDA fallback; and
- QPR, our query-assisted estimator for cumulative all-seen transduction, evaluated with identity-disjoint calibration, query self-exclusion, and explicit cohort and base–novel trade-off checks.

Together, the results show that scarce-class estimation is often more consequential than adding classifier flexibility, while also making clear that unlabeled-query repair is batch-dependent and not uniformly beneficial.

2. Related Work

FSCIL with fixed or adapted representations. FSCIL methods commonly modify the representation or classifier to preserve old classes and accommodate scarce new ones: CEC evolves classifiers, FACT reserves forward-compatible space, and NC-FSCIL, OrCo, and ASP impose geometric or prompt-based structure [1, 15, 37, 38, 40]. With strong pretrained features, however, simple fixed-feature heads can be competitive [34]. FSA limits adaptation to the first session, while RanPAC and RanDumb use analytic heads over fixed feature expansions [19, 21, 23]. We freeze the encoder in every session and isolate uncertainty in the resulting head.

Support-only estimation. Cosine NCM and prototypical networks represent a class by its support mean [20, 25]. TEEN, Distribution Calibration, and higher-order calibration transfer base-class structure to scarce classes [6, 31, 36]; EPT instead learns prototype offsets [10]. FeCAM and HyCal use covariance-aware distances, whereas ULA adjusts base–novel logits [5, 13, 33]. Distribution Calibration synthesizes features, higher-order calibration combines novel and semantically related base covariances after base-session adapter tuning, and EPT optimizes learned offsets. These are relevant calibration mechanisms but have different adaptation or synthetic data budgets from our fixed-feature closed-form head. ShrinkSel does not propose a new covariance estimator: it selects prototype radius jointly with classical covariance shrinkage [12], using class-disjoint base episodes and retaining cosine and shared LDA as exact endpoints.

Different modality budgets. PriViLege, FDR, BiMC, TIM++, and Transductive CLIP use text, prompts, language-derived initialization, or vision–text probability features [3, 14, 18, 22, 35]. They provide important multimodal context but are not information-matched to our frozen image-only heads; the primary rankings therefore keep them separate.

Transductive inference. TIM optimizes mutual information, LaplacianShot enforces graph smoothness, PT-MAP uses optimal transport, and ProtoLP alternates propagation with prototype refinement [2, 9, 41, 42]. These methods are usually studied in balanced, low-way, all-novel episodes and can depend on assumed query proportions [28]. Prototype Rectification also uses unlabeled queries, combining label propagation and feature shifting in episodic few-shot learning [16]; we therefore do not claim the first use of unlabeled queries for prototype rectification. Our QPR targets the asymmetric all-seen FSCIL setting: many-shot base prototypes remain fixed, only gated novel directions move, the support-selected radius and metric are preserved, and no class marginal is provided.

Diagnosis. Approximation-risk analysis studies transfer, margins, and base–novel balance in FSCIL [32]. Our narrower question begins after the representation is fixed. An ordered counterfactual changes novel means and then classifier form on the same features, separating an estimation-sensitive term from a residual without claiming a unique causal decomposition.

3. Diagnosing and Correcting Prototype Error

3.1. Setting and information regimes

Session 0 provides labeled base classes Y_0; session $t > 0$ adds a disjoint set Y_t with K labeled examples per class. Prediction after session t covers $\mathcal{C}_t = \bigcup_{j=0}^{t} Y_j$. A frozen image encoder produces

$$z(x) = f_{\text{img}}(x)/\|f_{\text{img}}(x)\|_2 \in \mathbb{R}^D. \tag{1}$$

Our CLIP pipeline calls only the image tower. We call Y_t, $t > 0$, *head-novel*: these identities are new to the downstream classifier, but their absence from encoder pretraining cannot generally be verified.

A *support-only* rule may use $z(x)$, labeled supports, and base statistics. A *query-assisted* rule may also use an unlabeled cohort $\mathcal{U}_t$ containing only classes in $\mathcal{C}_t$ and is therefore batch-dependent. Complete labeled novel data are restricted to the retrospective diagnosis. "Training-free" below means no gradient-based parameter fitting; closed-form base statistics and base-only selection remain allowed. Known query-class proportions are not evaluated as a deployable method.

3.2. Ordered matched counterfactual path

For support features S_c^K and the complete training split T_c, define

$$\hat{\mu}_c^K = \text{norm}\left(\sum_{z \in S_c^K} z\right), \qquad \hat{\mu}_c^{\text{full}} = \text{norm}\left(\sum_{z \in T_c} z\right), \tag{2}$$

where $\text{norm}(v) = v/\|v\|_2$ for $v \neq 0$. The complete-split mean is a finite labeled reference, not a population oracle. At the final session, A_n^K and A_n^{full} are class-balanced novel accuracies from cosine NCM with base means fixed and only

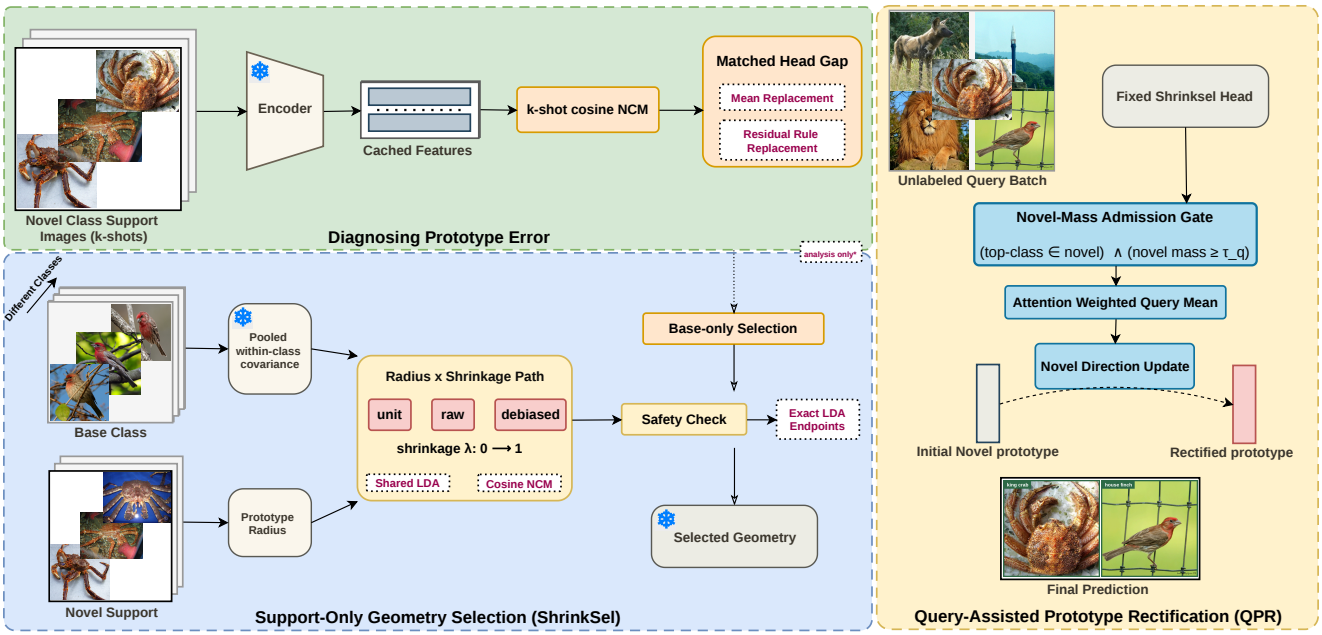

Figure 1. The analysis-only path first replaces five-shot novel means under fixed cosine scoring and then fits a complete-data linear probe. ShrinkSel selects support-only geometry from base classes; QPR adds a gated unlabeled-cohort update. Neither prediction path uses query labels or known class proportions.

novel means changed. Let A_n^{LP} be the accuracy of a fixed ℓ_2-regularized multinomial probe trained on the same complete features for all seen classes. The ordered path obeys

$$\underbrace{A_n^{\mathrm{LP}} - A_n^K}_{\text{matched head gap}} = \underbrace{A_n^{\mathrm{full}} - A_n^K}_{\text{mean replacement}} + \underbrace{A_n^{\mathrm{LP}} - A_n^{\mathrm{full}}}_{\text{residual rule change}}. \quad (3)$$

This is an exact accounting identity, not a unique causal decomposition. We report the ordered share $r = (A_n^{\mathrm{full}} - A_n^K)/(A_n^{\mathrm{LP}} - A_n^K)$ only when a crossed support-by-class bootstrap interval excludes a zero denominator. Probe, bootstrap, repair, and classwise definitions are in the supplement.

3.3. Finite-shot motivation

For an unnormalized mean $\hat{\mu}_c = \mu_c + \epsilon$ with $\mathbb{E}[\epsilon] = 0$, $\mathrm{Cov}(\epsilon) = \Sigma/K$, and ϵ independent of the query feature x, let G be a fixed positive-definite metric and $d_G(a,b) = (a-b)^\top G(a-b)$. Then

$$\mathbb{E}[d_G(x,\hat{\mu}_c) - d_G(x,\mu_c)] = \mathrm{tr}(G\Sigma)/K. \quad (4)$$

This states only an expected correct-class distance penalty. It is neither an accuracy bound nor a model of normalized prototypes, competing classes, or estimated G. Because G also changes between-class separation, it motivates selecting the discrimination–noise trade-off rather than prescribing a shrinkage value.

3.4. Support-only geometry selection

ShrinkSel estimates the pooled within-class covariance from complete base features,

$$\begin{aligned} \bar{z}_c^B &= |T_c|^{-1} \sum_{z \in T_c} z, \\ \bar{\Sigma}_B &= \frac{1}{M} \sum_{c \in Y_0} \sum_{z \in T_c} (z - \bar{z}_c^B)(z - \bar{z}_c^B)^\top, \\ \delta &= \max\{\mathrm{tr}(\bar{\Sigma}_B)/D, 10^{-12}\}, \end{aligned} \quad (5)$$

where T_c is the complete base training set and $M = \sum_c (|T_c| - 1)$. This covariance is frozen after the base session.

For n i.i.d. unit support features with sample mean $\bar{z}_c$ and population mean μ_c, the raw resultant length is upward biased because

$$\mathbb{E}\|\bar{z}_c\|_2^2 = \frac{1}{n} + \frac{n-1}{n}\|\mu_c\|_2^2. \quad (6)$$

For $n \geq 2$, we therefore use

$$\hat{r}_c = \sqrt{\left[\frac{n\|\bar{z}_c\|_2^2 - n^{-1}\sum_i \|z_i\|_2^2}{n-1}\right]_+}; \qquad \hat{r}_c = \|\bar{z}_c\|_2 \text{ if } n = 1. \quad (7)$$

Here $[a]_+ = \max(a, 0)$. Clipping ensures a valid radius but can reintroduce finite-sample bias. The prototype family is unit $u_c = \mathrm{norm}(\bar{z}_c)$, raw $\bar{z}_c$, or corrected $\hat{r}_c u_c$. Each is

crossed with

$$G_\lambda = [(1-\lambda)\bar{\Sigma}_B + (\lambda+\gamma)\delta I]^{-1}, \quad \gamma = 1,$$
$$s_c(z) = -\tfrac{1}{2}(z-p_c)^\top G_\lambda (z-p_c), \tag{8}$$
$$\lambda \in \{0, .1, .25, .5, .75, .9, 1\}.$$

Unit prototypes at $\lambda = 1$ reproduce cosine NCM, while raw prototypes at $\lambda = 0$ reproduce ridge-regularized shared-covariance LDA with equal class priors.

Base-only selection. Let $B = |Y_0|$, let each of T increments contain m classes, and define the class-count weights

$$w_n = \frac{1}{T+1}\sum_{t=0}^{T}\frac{tm}{B+tm}, \qquad w_b = 1 - w_n. \tag{9}$$

For candidate $h = (p, \lambda)$, each pseudo-increment measures anchor and pseudo-novel accuracy, $A_b(h)$ and $A_n(h)$, and scores $J(h) = w_b A_b(h) + w_n A_n(h)$. Base identities are first split into construction, selection, and safety partitions. For ShrinkSel, construction and selection identities are pooled and deterministically repartitioned into four class-disjoint pseudo-novel folds; safety identities never enter selection. Across 200 episodes, a candidate is feasible only if it does not trail raw $\lambda = 0$ LDA in any fold. The feasible candidate with highest pooled J is checked on safety; a regression beyond 0.10 point deploys the exact LDA endpoint. The chosen geometry is refit on authorized base features and frozen. Split sizes, seeds, episode construction, tie-breaking, and the cumulative-count stress test are detailed in the supplement.

3.5. Query-assisted prototype rectification

Query Prototype Rectification (QPR) is introduced here and operates on the fixed ShrinkSel head. Initial scores define

$$\pi(c \mid z) = \mathrm{softmax}_{c\in\mathcal{C}_t}(\kappa s_c(z)), \quad m_n(z) = \sum_{c\in\mathcal{C}_t\setminus Y_0}\pi(c \mid z). \tag{10}$$

A novel class c admits

$$\begin{aligned}\mathcal{G}_c = \{z \in \mathcal{U}_t :\ & \arg\max_{d\in\mathcal{C}_t}\pi(d \mid z) \notin Y_0,\\ & m_n(z) \ge \tau_q,\\ & c = \arg\max_{d\in\mathcal{C}_t\setminus Y_0}\pi(d \mid z)\}.\end{aligned} \tag{11}$$

If $|\mathcal{G}_c| < m_{\min}$, its prototype is unchanged. Otherwise, define $a_c(z) = \pi(c \mid z)/\sum_{u\in\mathcal{G}_c}\pi(c \mid u)$ and perform

$$\begin{aligned}\tilde{\mu}_c &= \mathrm{norm}\left(\sum_{z\in\mathcal{G}_c} a_c(z) z\right),\\ v_c' &= \mathrm{norm}((1-\alpha)\,\mathrm{norm}(p_c) + \alpha\tilde{\mu}_c),\\ p_c' &= \|p_c\|_2 v_c'.\end{aligned} \tag{12}$$

Base prototypes, non-admitted novel prototypes, and G_λ remain fixed; all queries are then rescored by Eq. (8). The update is one-shot, restarts from the support-only head each session, and assumes neither labels nor class proportions. Ordinary QPR uses the same cohort to construct and evaluate the update. Exact self-exclusion subtracts query z_i from its admitted aggregate when classifying z_i; other cohort queries remain available, so this is still transductive rather than online inference.

The four controls are selected on identity-disjoint base pseudo-query episodes spanning balanced and deterministic-imbalanced cohorts, with held-out combined and anchor safety gates; failure deploys the exact no-update case $\alpha = 0$. The 96-point grid, gate thresholds, calibration episodes, tie-breaking, vectorized self-exclusion, and cumulative-competition calibration are detailed in the supplement.

4. Experiments

4.1. Experimental design

We follow BiMC's five-shot class orders and manifests [3]. CUB-200 uses $100 + 10 \times 10$ classes; CIFAR-100 and mini-ImageNet use $60 + 8 \times 5$ [11, 29, 30]. Aircraft and DTD provide additional domain checks [4, 17]. Results average twenty matched support draws; the five-shot supports are nested in a $K \in \{1, 2, 5, 10, 20\}$ study.

The primary matrix uses frozen OpenAI CLIP ViT-B/16, ViT-B/32, and RN50 image features [24]; MAE-ft ViT-B/16 and supervised ImageNet ResNet-50 test cross-family transfer [7, 8]. A ResNet-18 trained only on each benchmark's base classes is a weaker class-disjoint control. The released encoders do not establish pretraining novelty, so all claims concern classes new to the head.

Support-only comparisons include NCM, shared/diagonal LDA, TEEN, ULA, FeCAM, RanPAC, RanDumb, and an image-only reduction of HyCal [5, 13, 19, 20, 23, 31, 33]. Query-assisted comparisons use TIM-GD, LaplacianShot, ProtoLP, and uniform-marginal MAP-OT$_u$ [2, 9, 41, 42]. Every matched comparison shares features, supports, sessions, and queries. The primary metric is the unweighted mean of sample-weighted session accuracies, including session 0. Paired method intervals use 10,000 support resamples; the diagnostic crossed support–class intervals use 5,000 draws. Full metric, selection, provenance, and implementation details appear in the supplement.

4.2. Locating the recoverable error

The complete-mean endpoint passes half of the matched five-shot-to-probe gap in every row of Fig. 2. Because features, base prototypes, candidate classes, and scoring remain fixed along the blue step, this pattern specifically localizes most of the ordered gap to scarce novel statistics. It is not simply evidence that more labeled data improve an arbitrary head.

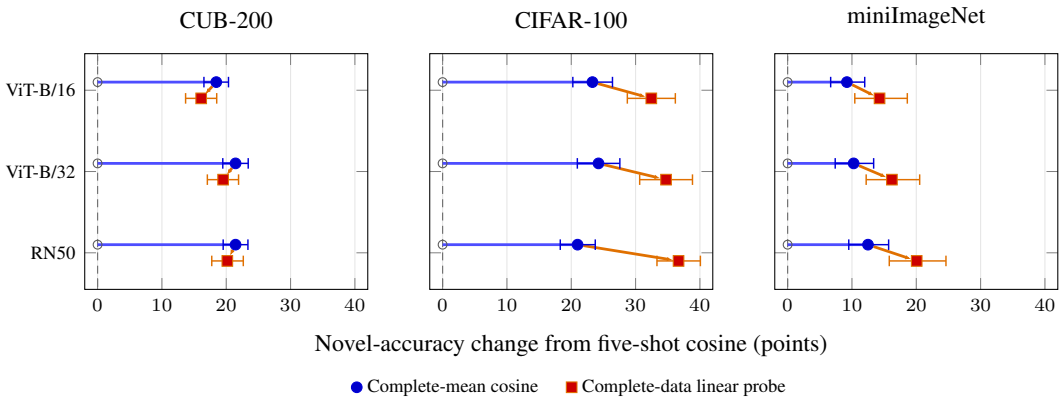

Figure 2. Ordered matched counterfactual path for three datasets and three CLIP encoders. Each row starts from five-shot cosine NCM at zero. The blue endpoint changes only novel means to complete-data means while preserving cosine scoring; the orange endpoint then changes to a linear probe trained on the same complete-data images. Arrows show this second ordered change, so a leftward arrow means that the probe underperforms complete-mean cosine. Whiskers are 95% two-way bootstrap intervals over supports and novel classes. The path is an accounting device, not a causal decomposition.

The same majority conclusion holds across the cross-family and Aircraft checks, giving sixteen strong-representation settings in total.

The direction of the orange step changes across datasets. Its leftward orientation on CUB shows that a full-data linear rule is not consistently better than full-data cosine prototypes. Extra classifier flexibility therefore cannot be treated as an oracle once estimation is controlled; its value depends on how a representation arranges within- and between-class variation. This is why Eq. (3) is an ordered description rather than a causal partition.

Three diagnostics in the supplement strengthen, but also delimit, this interpretation. Accuracy improves monotonically as nested supports reduce mean uncertainty, and complete means repair substantially more decisions than they harm. Prototype angular error is associated with classwise loss on the standard matrix, yet Aircraft breaks that classwise relationship despite a large aggregate replacement effect. Angle measures estimation error; whether it flips a label also depends on the nearest competing margin. The aggregate matched intervention is consequently the primary evidence, with shot and angle trends serving as mechanism checks.

4.3. Support-only recovery

Table 1 reports the detailed CUB-200 trajectory. The relevant endpoint is shared LDA, not cosine: isotropic cosine geometry leaves substantial structure unused, whereas LDA is already inside the ShrinkSel family. ShrinkSel is slightly below LDA in sessions 0–1 and then overtakes it as scarce classes accumulate, so its advantage is not an across-session dominance claim.

Figure 3 extends the trajectory comparison across datasets. ShrinkSel and shared LDA remain close early, then separate as scarce classes accumulate on CUB and miniImageNet, while CIFAR stays nearly tied. RanPAC's strong base endpoint does not persist as novel classes enter, showing why base-session strength alone does not determine cumulative all-seen performance.

The ablations explain why the selector helps. Covariance geometry supplies most of the improvement over cosine because it emphasizes directions that separate base classes. Prototype radius is a smaller, representation-dependent correction: the norm of a mean contains concentration information, but its few-shot estimate is inflated by Eq. (6). Keeping both choices coupled matters because a metric that magnifies low-variance directions can also magnify mean noise, exactly the trade-off suggested by Eq. (4). Consequently, selected shrinkage spans the grid rather than converging to a universal dataset-specific value.

Figure 4 makes the selector's two roles visible. CUB has a broad feasible ridge around moderate shrinkage, so several radius treatments induce nearly equivalent geometry. CIFAR instead eliminates most shrinkage choices because at least one class-disjoint fold regresses, while miniImageNet retains a short plateau among corrected-radius choices near the covariance endpoint. The selected cell is therefore determined jointly by mean performance and worst-fold safety, not by a smooth universal optimum. Target-oracle and leave-one-fold-out diagnostics in the supplement reach the same conclusion: discrete parameters can change while accuracy barely does. We claim transfer of audited performance, not recovery of a unique shrinkage coefficient. Novel-order permutations retain the qualitative result but do not vary the base/novel identity split.

The supplement also compares matched image-only adapters and joint probes. Ordinary sample-weighted opti-

Table 1. Support-only session-wise accuracy on CUB-200 with frozen CLIP ViT-B/16. Sessions 1–10 average twenty matched support draws; session 0 is deterministic. Avg. is the unweighted session average and PD is session-0 minus final-session accuracy. Under frozen features, PD measures cumulative class competition rather than encoder forgetting. Bold and underline mark the best and second-best displayed entries. †HyCal is our image-only algebraic hard-threshold reduction, not general HyCal.

Method	Accuracy in each session (%) ↑											Avg. ↑	PD ↓
	0	1	2	3	4	5	6	7	8	9	10		
Cosine-NCM	81.63	79.14	77.49	74.25	73.26	71.69	70.81	70.04	68.53	68.72	68.48	73.09	13.16
TEEN (released)	81.63	78.54	76.64	72.90	71.85	70.33	69.50	68.68	67.08	67.26	67.09	71.95	14.54
ULA (released)	81.63	79.18	77.57	74.30	73.20	71.63	70.93	70.16	68.61	68.83	68.62	73.15	13.01
FeCAM	77.06	71.92	67.35	63.21	60.64	58.29	55.68	53.69	51.68	50.81	49.71	60.00	27.35
HyCal image-only reduction†	79.78	74.06	69.27	64.92	62.10	59.66	56.85	54.87	52.70	51.95	50.80	61.54	28.98
RanDumb (RBF-NCM)	81.56	78.96	77.21	74.02	72.93	71.35	70.47	69.70	68.22	68.35	68.08	72.80	13.49
RanPAC (10k projection)	83.62	79.48	76.09	72.65	70.96	68.77	66.86	65.71	63.86	63.75	63.09	70.44	20.54
Shared-covariance LDA	**84.88**	**81.85**	80.03	77.12	76.23	74.70	73.80	73.34	72.14	72.32	72.20	76.24	12.68
ShrinkSel	84.25	81.71	**80.23**	**77.36**	**76.50**	**75.08**	**74.38**	**73.87**	**72.62**	**72.83**	**72.72**	**76.50**	**11.53**

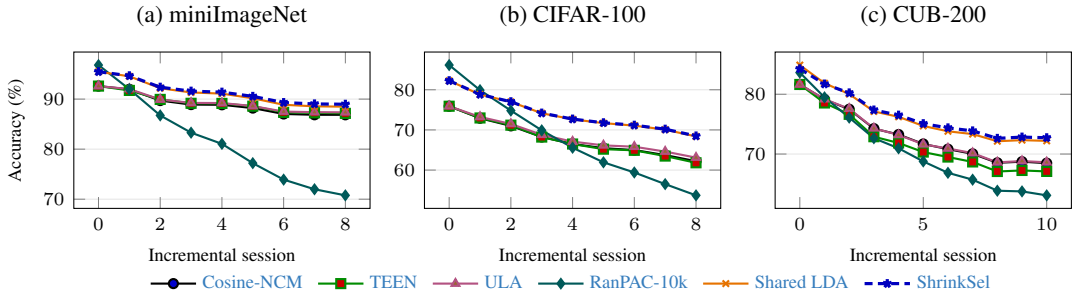

Figure 3. Support-only session-wise accuracy on the three standard benchmarks with frozen CLIP ViT-B/16. Curves are means over twenty matched support draws; session 0 is deterministic. The six displayed methods keep the comparison legible, while the supplement reports the complete matrices. Panels use dataset-specific vertical ranges.

mization is dominated by abundant replayed base examples, so it can strengthen the base block while leaving the five-example novel estimate poor. Inverse-frequency weighting shifts this balance and can reverse the overall ranking when base performance carries sufficient sample weight, but it still does not remove the novel-data bottleneck uniformly. These controls show why "optimized" is not an information-matched upper bound on a closed-form scarce-class head.

4.4. What an unlabeled cohort adds

Table 2 separates two questions that the former absolute-only ranking conflated. QPR on ShrinkSel is the best complete tested pipeline on all three datasets, but part of that ranking comes from its stronger support-only head. Relative to each pipeline's declared support-only reference, QPR has the largest gain on CUB and CIFAR, while TIM-GD's gain over cosine is slightly larger on miniImageNet. We therefore claim the best complete tested pipeline, not universal operator superiority. Exact self-exclusion retains positive gains on all three datasets and improves CUB relative to same-cohort QPR; the benefit thus comes from peer queries rather than a query reinforcing its own prediction.

The source transductive methods were designed mainly for balanced low-way episodes in which every candidate class is novel. Cumulative FSCIL instead combines many well-estimated base classes with a growing scarce-class block. A separate native five-way check confirms that TIM-GD improves in its customary regime, so its weaker all-seen behavior is a protocol-transfer issue rather than implementation collapse. QPR is tailored to this asymmetry: it gates for novel-like mass, leaves base prototypes fixed, and preserves the support-selected radius. This reduces novel direction error, but moving a novel boundary can also capture base queries. The resulting gain is therefore a novel-repair/base-damage trade-off, not a uniformly safer classifier.

Component diagnostics support controlled direction repair rather than fine posterior weighting. Re-estimating radius from pseudo-labeled queries is consistently harmful: it changes both direction and the base–novel score scale. Removing all conservative controls also degrades the operator, whereas uniform within-group weights have little effect. Individual gates can be nearly inactive on a complete cohort,

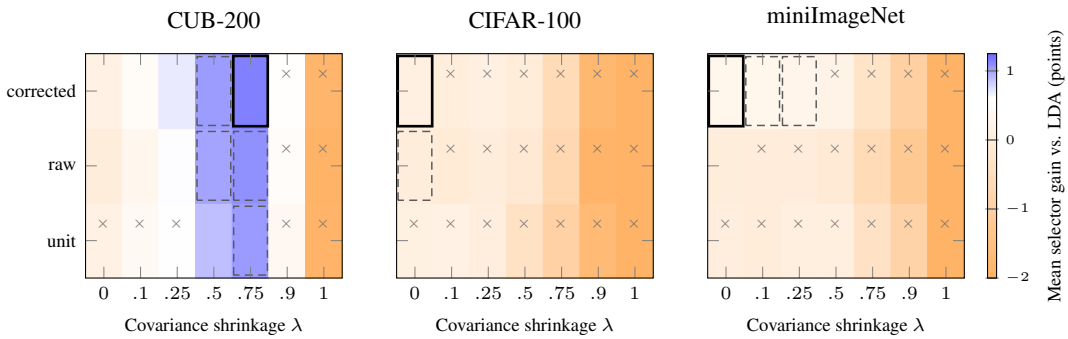

Figure 4. The complete ViT-B/16 ShrinkSel selection landscape, computed only from class-disjoint base folds. Color encodes the mean schedule-weighted gain over shared LDA, clipped at -2 points to keep the feasible region legible; $\times$ marks candidates rejected by the non-regression constraint. The solid box is selected, and dashed boxes are other feasible candidates within 0.15 point of its objective.

Table 2. Matched query-assisted comparison with frozen CLIP ViT-B/16. Avg. is complete-pipeline session-average accuracy; Δ Ref. is the paired change from the method's declared support-only reference, with a 95% bootstrap interval over twenty matched support draws. QPR uses ShrinkSel as its reference; all transferred comparators use Cosine-NCM. Bold and underline independently mark the largest and second-largest Avg. and Δ Ref. in each dataset using unrounded values. MAP-OT$_u$ additionally assumes a uniform query marginal.

Method	Ref.	CUB-200		CIFAR-100		miniImageNet	
		Avg. ↑	Δ Ref. [95%] ↑	Avg. ↑	Δ Ref. [95%] ↑	Avg. ↑	Δ Ref. [95%] ↑
ShrinkSel + QPR	ShrinkSel	<u>78.13</u>	<u>+1.63 [+1.53, +1.71]</u>	**76.10**	**+2.03 [+1.93, +2.15]**	**92.69**	<u>+1.21 [+1.15, +1.28]</u>
ShrinkSel + QPR (self-excl.)	ShrinkSel	**78.35**	**+1.84 [+1.73, +1.94]**	<u>76.10</u>	<u>+2.03 [+1.91, +2.14]</u>	<u>92.67</u>	+1.19 [+1.12, +1.26]
TIM-GD	Cosine-NCM	72.99	-0.11 [-0.22, 0.00]	69.40	+1.51 [+1.43, +1.60]	90.33	**+1.33 [+1.28, +1.38]**
LapShot-full	Cosine-NCM	69.71	-3.38 [-3.49, -3.27]	63.18	-4.71 [-4.84, -4.57]	86.02	-2.98 [-3.03, -2.92]
LapShot-core	Cosine-NCM	73.03	-0.06 [-0.07, -0.06]	67.80	-0.08 [-0.09, -0.07]	89.03	+0.03 [+0.02, +0.04]
ProtoLP	Cosine-NCM	66.90	-6.20 [-6.43, -5.97]	62.75	-5.14 [-5.24, -5.03]	86.71	-2.28 [-2.38, -2.18]
MAP-OT$_u$	Cosine-NCM	58.13	-14.96 [-15.11, -14.80]	59.35	-8.53 [-8.64, -8.42]	78.35	-10.65 [-10.84, -10.46]

so the ablation does not imply that every gate is necessary on every dataset; it shows that preserving the parent geometry and avoiding an unconstrained update are the important design choices.

The one-dimensional sweeps in Fig. 5 rule out two simple stories. CUB and CIFAR peak at two supports, whereas miniImageNet is effectively tied across one and two: extreme support scarcity creates repair headroom but can also weaken the assignments that drive rectification. Past that balance, increasing K steadily removes the estimation error QPR can repair. Along the cohort axis, the minimum-count gate usually makes one- and two-query cohorts a no-op; a barely activated update can be adverse before ten or twenty peer queries make the admitted direction reliable.

The exploratory crossed surface in Fig. 6 shows that these are interacting error sources rather than independent monotone benefits. Large cohorts are most useful while the support estimate is still noisy, whereas their advantage contracts as labeled means stabilize. Conversely, the intermediate five-query band can move a prototype using too little reliable peer evidence, producing the clearest adverse region on CUB. This transition explains both the non-monotone support optimum and why QPR cannot be summarized as a fixed bonus for consuming queries. Architecture extensions retain the pattern within CLIP, and DTD shows that QPR can improve a fixed parent even when ShrinkSel itself falls below LDA. The evaluated datasets and CLIP backbones establish empirical scope, not universal cross-domain or cross-family generalization.

4.5. Boundary and cost checks

The base-only ResNet-18 control does not pass the majority criterion on any standard dataset. Because representation strength and pretraining exposure change together, this bounds the diagnosis without identifying which factor is responsible. Resource profiling on cached ViT-B/16 features shows that ShrinkSel stores one pooled eigensystem and class prototypes and adds novel means in closed form. These numbers, reported in the supplement, support a lightweight-head claim after feature extraction, not an end-to-end latency claim. QPR remains cohort-dependent regardless of its small head state.

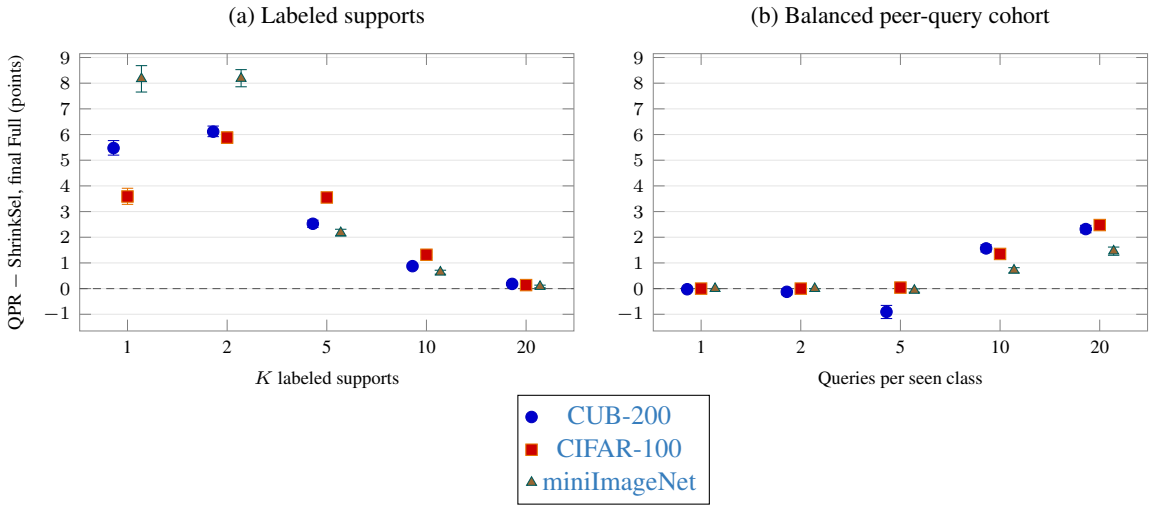

Figure 5. QPR operating boundary on ViT-B/16. The left panel varies labeled support under the complete all-seen cohort; the right fixes $K = 5$ and varies a controlled balanced cohort. Both report final-session Full-accuracy gain over the same ShrinkSel parent, with paired 95% bootstrap intervals over twenty supports. The QPR tuple is fixed throughout.

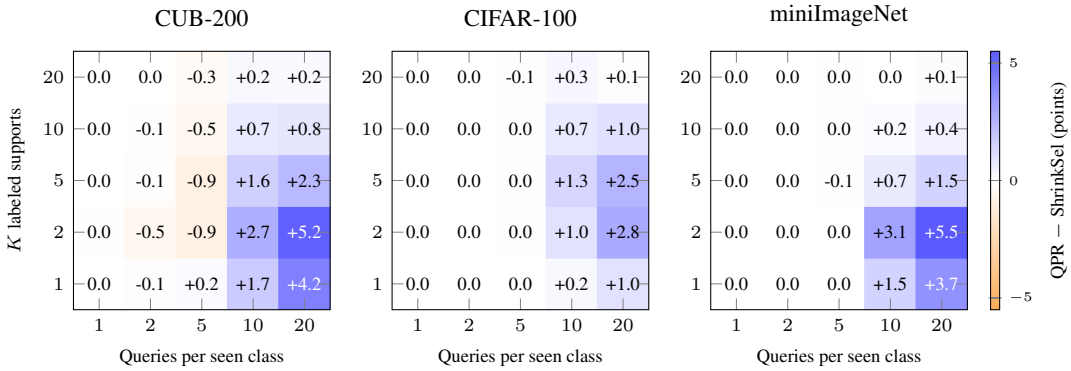

Figure 6. Exploratory interaction between labeled support and balanced unlabeled-cohort size at the final session. Each cell is QPR minus ShrinkSel Full accuracy in points. Labels construct the controlled cohort subset but are hidden from both methods. The five-shot-selected QPR tuple is applied to every cell without reselection; no cellwise significance claim is made over this 75-condition surface.

5. Limitations

Our released encoders establish novelty only at the incremental head. CLIP's corpus cannot be audited, and the additional encoder families use ImageNet supervision. The base-only ResNet control excludes novel benchmark classes but is also weaker, so it cannot separate exposure from representation quality.

The complete-data means and linear probe are retrospective references. Their accounting is exact for the chosen order, but the two terms are descriptive rather than causal. ShrinkSel is a conservative combination of classical estimators; some gains over LDA are practically small, its selector only approximates cumulative novel competition, and transfer is conditioned on a fixed base/novel identity split.

QPR is batch transductive, not an online or support-only method. Other cohort queries remain available even under query-wise self-exclusion, and moving novel prototypes can harm base predictions.

6. Conclusion

Matched counterfactuals across strong frozen representations identify novel-mean replacement as the larger component of the tested ordered head gap. This diagnosis motivates two deliberately bounded corrections: ShrinkSel selects support-only radius and covariance geometry with an exact LDA fallback, while the new QPR method uses an unlabeled cohort to repair novel directions without changing that geometry. Their behavior shows that frozen-feature FSCIL is governed less by a universally superior classifier than by which evidence is available to estimate each scarce class reliably.

References

[1] Noor Ahmed, Anna Kukleva, and Bernt Schiele. Orco: Towards better generalization via orthogonality and contrast for few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 28762–28771, 2024. 2

[2] Malik Boudiaf, Ziko Imtiaz Masud, Jérôme Rony, Jose Dolz, Pablo Piantanida, and Ismail Ben Ayed. Transductive information maximization for few-shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 2, 4

[3] Yiyang Chen, Tianyu Ding, Lei Wang, Jing Huo, Yang Gao, and Wenbin Li. Enhancing few-shot class-incremental learning via training-free bi-level modality calibration. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9881–9890, 2025. 2, 4

[4] Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3606–3613, 2014. 4

[5] Dipam Goswami, Albin Soutif-Cormerais, Yuyang Liu, Sandesh Kamath, Bartłomiej Twardowski, and Joost van de Weijer. FeCAM: Exploiting the heterogeneity of class distributions in exemplar-free class-incremental learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2, 4

[6] Dipam Goswami, Bartłomiej Twardowski, and Joost Van De Weijer. Calibrating higher-order statistics for few-shot class-incremental learning with pre-trained vision transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 4075–4084, 2024. 2

[7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016. 4

[8] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009, 2022. 4

[9] Yuqing Hu, Vincent Gripon, and Stéphane Pateux. Leveraging the feature distribution in transfer-based few-shot learning. In *International Conference on Artificial Neural Networks (ICANN)*, pages 487–499, 2021. 2, 4

[10] Shengqin Jiang, Xiaoran Feng, Yuankai Qi, Haokui Zhang, Renlong Hang, Qingshan Liu, Lina Yao, Quan Z. Sheng, and Ming-Hsuan Yang. Unlocking prototype potential: An efficient tuning framework for few-shot class-incremental learning. *arXiv preprint arXiv:2602.05271*, 2026. 2

[11] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. 4

[12] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. 2

[13] Eunju Lee, MiHyeon Kim, JuneHyoung Kwon, Yoonji Lee, JiHyun Kim, Soojin Jang, and YoungBin Kim. HyCal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 29462–29471, 2026. 2, 4

[14] Yingping Li, Yutong Zou, Yunshi Huang, Changzhe Jiao, Xinlin Wang, Shen Peng, Zhang Guo, and Shuiping Gou. TIM++: Transductive information maximization for few-shot CLIP. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6671–6680, 2026. 2

[15] Chenxi Liu, Zhenyi Wang, Tianyi Xiong, Ruibo Chen, Yihan Wu, Junfeng Guo, and Heng Huang. Few-shot class incremental learning with attention-aware self-adaptive prompt. In *European Conference on Computer Vision (ECCV)*, pages 1–18, 2024. 2

[16] Jinlu Liu, Liang Song, and Yongqiang Qin. Prototype rectification for few-shot learning. In *European Conference on Computer Vision (ECCV)*, pages 741–756, 2020. 2

[17] Subhransu Maji, Juho Kannala, Esa Rahtu, Matthew Blaschko, and Andrea Vedaldi. Fine-grained visual classification of aircraft. Technical Report 1306.5151, arXiv, 2013. 4

[18] Ségolène Martin, Yunshi Huang, Fereshteh Shakeri, Jean-Christophe Pesquet, and Ismail Ben Ayed. Transductive zero-shot and few-shot CLIP. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 28816–28826, 2024. 2

[19] Mark D McDonnell, Dong Gong, Amin Parvaneh, Ehsan Abbasnejad, and Anton van den Hengel. RanPAC: Random projections and pre-trained models for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2, 4

[20] Thomas Mensink, Jakob Verbeek, Florent Perronnin, and Gabriela Csurka. Distance-based image classification: Generalizing to new classes at near-zero cost. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 35 (11):2624–2637, 2013. 2, 4

[21] Aristeidis Panos, Yuriko Kobe, Daniel Olmeda Reino, Rahaf Aljundi, and Richard E Turner. First session adaptation: A strong replay-free baseline for class-incremental learning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 2

[22] Keon-Hee Park, Kyungwoo Song, and Gyeong-Moon Park. Pre-trained vision and language transformers are few-shot incremental learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23881–23890, 2024. 2

[23] Ameya Prabhu, Shiven Sinha, Ponnurangam Kumaraguru, Philip H. S. Torr, Ozan Sener, and Puneet K. Dokania. RanDumb: Random representations outperform online continually learned representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. 2, 4

[24] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021. 4

[25] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. 2

[26] Zeyin Song, Yifan Zhao, Yujun Shi, Peixi Peng, Li Yuan, and Yonghong Tian. Learning with fantasy: Semantic-aware virtual contrastive constraint for few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24183–24192, 2023. 1

[27] Xiaoyu Tao, Xiaopeng Hong, Xinyuan Chang, Songlin Dong, Xing Wei, and Yihong Gong. Few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12183–12192, 2020. 1

[28] Olivier Veilleux, Malik Boudiaf, Pablo Piantanida, and Ismail Ben Ayed. Realistic evaluation of transductive few-shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 9290–9302, 2021. 2

[29] Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Daan Wierstra, et al. Matching networks for one shot learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. 4

[30] Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The Caltech-UCSD Birds-200-2011 dataset. Technical Report CNS-TR-2011-001, California Institute of Technology, 2011. 4

[31] Qi-Wei Wang, Da-Wei Zhou, Yi-Kai Zhang, De-Chuan Zhan, and Han-Jia Ye. Few-shot class-incremental learning via training-free prototype calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2, 4

[32] Xuan Wang, Zhong Ji, Xiyao Liu, Yanwei Pang, and Jungong Han. On the approximation risk of few-shot class-incremental learning. In *European Conference on Computer Vision (ECCV)*, pages 162–178, 2024. 2

[33] Sungwon Woo, Dongjun Hwang, Shiwon Kim, Junsuk Choe, and Jongho Nang. Training-free uncertainty-guided logit adjustment for few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings*, pages 7634–7643, 2026. 2, 4

[34] Tz-Ying Wu, Gurumurthy Swaminathan, Zhizhong Li, Avinash Ravichandran, Nuno Vasconcelos, Rahul Bhotika, and Stefano Soatto. Class-incremental learning with strong pre-trained models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9601–9610, 2022. 2

[35] Zongyao Xue, Meina Kan, Shiguang Shan, and Xilin Chen. Feature decomposition-recomposition in large vision-language model for few-shot class-incremental learning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3153–3162, 2025. 2

[36] Shuo Yang, Lu Liu, and Min Xu. Free lunch for few-shot learning: Distribution calibration. In *International Conference on Learning Representations (ICLR)*, 2021. 2

[37] Yibo Yang, Haobo Yuan, Xiangtai Li, Zhouchen Lin, Philip Torr, and Dacheng Tao. Neural collapse inspired feature-classifier alignment for few-shot class-incremental learning. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2

[38] Chi Zhang, Nan Song, Guosheng Lin, Yun Zheng, Pan Pan, and Yinghui Xu. Few-shot incremental learning with continually evolved classifiers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12455–12464, 2021. 1, 2

[39] Jinghua Zhang, Li Liu, Olli Silvén, Matti Pietikäinen, and Dewen Hu. Few-shot class-incremental learning for classification and object detection: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(4):2924–2945, 2025. 1

[40] Da-Wei Zhou, Fu-Yun Wang, Han-Jia Ye, Liang Ma, Shiliang Pu, and De-Chuan Zhan. Forward compatible few-shot class-incremental learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9046–9056, 2022. 1, 2

[41] Hao Zhu and Piotr Koniusz. Transductive few-shot learning with prototype-based label propagation by iterative graph refinement. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23996–24006, 2023. 2, 4

[42] Imtiaz Masud Ziko, Jose Dolz, Eric Granger, and Ismail Ben Ayed. Laplacian regularized few-shot learning. In *International Conference on Machine Learning (ICML)*, pages 11660–11670, 2020. 2, 4